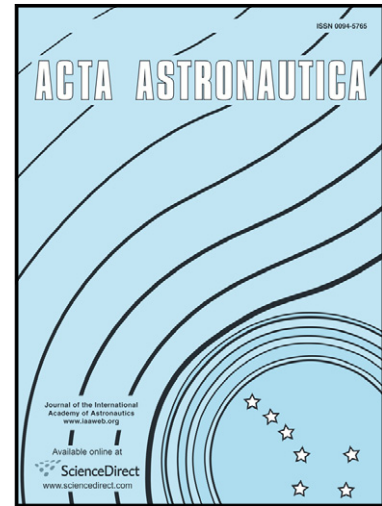


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Triple Shock Configurations with Negative Angle of Reflection

L.G. Gvozdeva¹, M.V. Silnikov^{2,3*}, S.A. Gavrenkov¹

¹ Joint Institute for High Temperatures of the Russian Academy of Sciences,
13-2 Izhorskaya Str., 125412 Moscow Russia

² Special Material Corp., 28A Bolshoy Sampsonievsky Ave., 194044 St. Petersburg Russia

³ Peter the Great Saint Petersburg Polytechnic University,
29 Polytechnicheskaya Str., 195251 St. Petersburg Russia

* Corresponding author: e-mail director@npo-sm.ru, tel./fax +78122729216

Summary

A triple configuration of stationary shocks in a steady gas flow is studied analytically. The study reveals that a new pattern of the triple-shock configuration is formed within the range of Mach numbers exceeding 3.0 and specific heat ratios below 1.4. The reflected wave coming from the triple point deviates down and forms a negative angle of reflection with the initial flow direction. This shock-wave pattern differs from the known positive reflection angle configuration where the reflected wave is inclined upwards to the initial flow direction. The negative angle configuration effect is considered in this study in the problem of a supersonic flow about two wedges. It has been shown, that this configuration is unstable and its appearance leads to a radical change of the total flow pattern. It is suggested that this effect can cause an evident alteration of the flow wave pattern in the problem of a supersonic jet nozzle structure.

The range of flow parameters in this study (large Mach numbers and small ratios of gas specific heats) corresponds to spacecraft flights (for example, in Mars atmosphere), as well as flows from the rocket nozzles with triple configurations at Mach reflections in both overexpanded and underexpanded jets. Instability supersonic flows with the analyzed triple configurations can influence both the aerodynamics of spacecraft flight and the functioning of its propulsion engine.

Keywords: triple configuration, negative reflection angle, flow instability.

Introduction

Shock wave reflection is a fundamental problem of internal and external aerodynamics [1-3]. One of the basic elements of a reflection pattern is a triple configuration.

Two basic types of reflection are known in a steady supersonic gas flow – that are regular and irregular (Mach) reflections. The triple configurations are formed at Mach reflection, for example, inside supersonic inlets or in overexpanded flows out from the space rocket engines [4-6]. A supersonic jet outflow from a nozzle occurs at the incalculable regimes of jet propulsions, various jet devices, for example ejectors, and in other cases, when a gas flow pressure at a nozzle exit differs from the ambient pressure. Gas flow over-expansion in the nozzle ($P_1 < P_2$) as well as flow under-expansion ($P_1 > P_2$) are possible, where P_1 is gas pressure at the nozzle exit section, P_2 is the surrounding pressure. As a result, various systems of alternating compression and expansion waves and shocks are formed in the jet outside the nozzle. A typical pattern of an overexpanded flow from a nozzle is shown in Fig. 1a (Fig. 1 is

analogous to flow scheme in [7]. The shock-wave configuration can be seen where IA and $I'A'$ are the incident shocks, AR and $A'R'$ are the reflected shocks, AM and $A'M$ are Mach stems. A similar pattern is formed in an air inlet at a supersonic velocity of an aircraft (Fig. 1b). In this case incident shock wave occurs as attached shocks at the wedges.

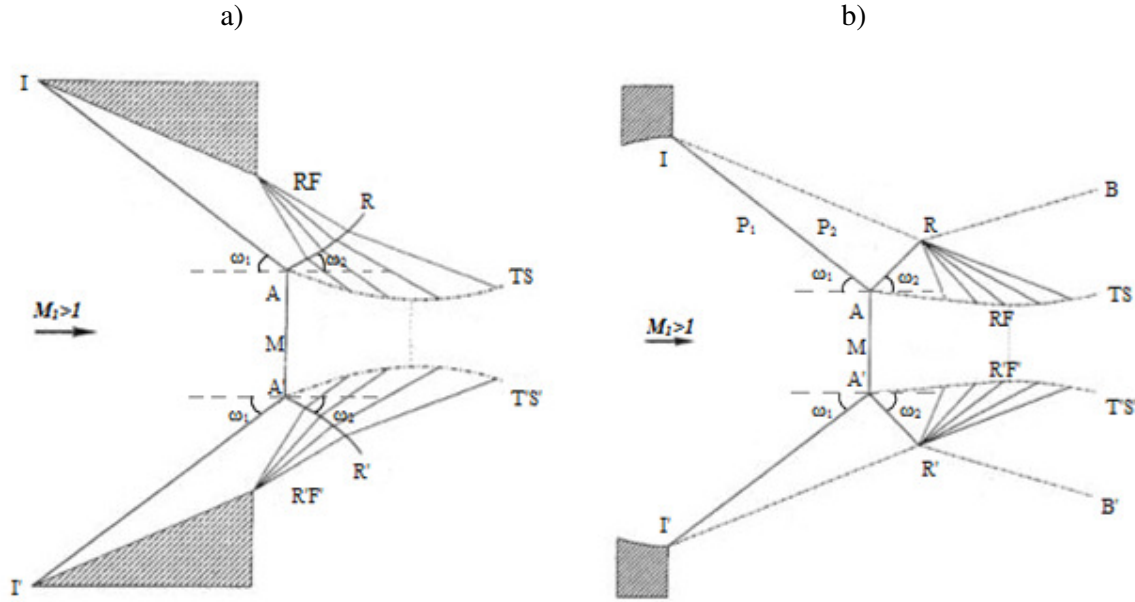


Fig. 1: a) overexpanded jet flow from the plane nozzle and formation of the triple-shock configuration; b) plane flow at the between two wedges modeling the supersonic inlet entrance and formation of the triple configuration at the channel. IA – incident wave, AR – reflected wave, ATS – slipstream, RF – rarefaction fan, IRB – jet boundaries.

The triple-wave configurations can be calculated independently of boundary problems of their origin using the triple point theory [8-13]. Let us consider a plane supersonic flow with gas specific heats ratio γ and March number M_1 (Fig. 2). It is assumed that a triple-shock system exists and transforms a uniform flow into two flows – the first of them propagates through incident (IA) and reflected (AR) shocks, and the second of them crosses Mach stem AM . Those flows are divided by the contact discontinuity (the slipstream) AT . It is also assumed that the conservation laws for shock waves (Hugoniot conditions) are fulfilled at each of the discontinuities; the consistency conditions are fulfilled at the slipstream (a flow propagating through an incident wave and a reflected wave is parallel to a flow downstream, through a Mach wave; pressure values at the both sides of the slipstream are equal). Shock strengths and slope angles in perfect gas depend on Mach number M_1 of an incoming flow, incident shock inclination angle ω_1 and the ratio γ of gas specific heats.

It is generally acknowledged in the problems of a shock reflection that the configurations presented in Fig. 2a (type II according to the classification in [12]) is typical. This conclusion is based on the calculations which deal with the specific heat ratio 1.4 primarily, because the steady reflections are usually studied experimentally in wind tunnels where a working gas is an air. The figures 1a and 1b show

the examples of such flows. In both triple configurations shown in Fig. 1 the reflected wave is situated in the same half-plane that the incident wave. Further we call such configuration “triple configuration with positive angle of reflection”.

Some dependencies of shock disposition in a triple configurations on the values of ω_1 and M_1 , and the specific heats ratio γ have been studied recently in [14-16]. New triple configuration with a negative reflection angle has been found (see Fig. 2b). Its distinctive feature is that the reflected wave is situated below the initial flow line (angle ω_2 is negative). Let us consider the conditions of formation of such a configuration and possible effects of its occurrence in the problems of shock waves reflection.

The results achieved in this study are relevant to overexpanded flows out off the space rocket engines, or rotating detonation wave engines flow structure in the zones of transverse waves interaction with the leading shock wave (see, for example, [17]). So they seem to be important in many aspects of rocket and spacecraft engineering.

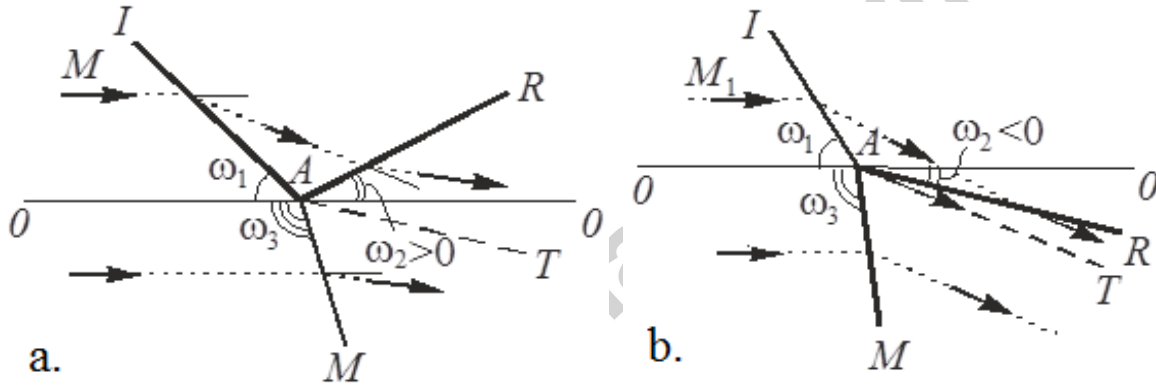


Fig. 2. Triple-shock configurations: a) at $\omega_2 > 0$; b) at $\omega_2 < 0$.

Theoretical Analysis of Triple Configurations

To define the region of flow regimes where triple configurations with negative reflection angles are generated, we use the analytical method of shock polars. Let us consider the steady supersonic gas flow with specific heats ratio γ and a Mach number M_1 (Fig. 3). The incident wave IA is inclined at an angle ω_1 to the initial flow direction. The gas flow behind the incident wave is deflected at an angle θ_1 . The gas stream behind the reflected wave is deflected at an angle θ_2 in the opposite direction. The Mach stem deflects the gas flow at an angle θ_3 . The flow behind the reflected shock must be parallel to the flow downstream the Mach stem. The pressures must be the same at both sides of the contact discontinuity. Therefore the compatibility conditions at the slipstream can be written as follows: $p_3 = p_4$, $\theta_3 = \theta_1 - \theta_2$.

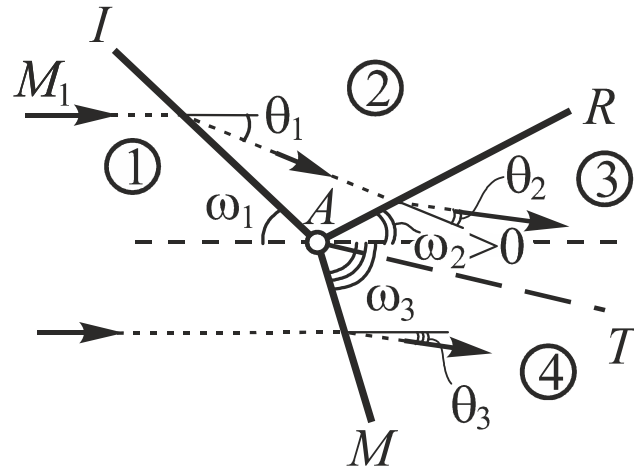


Fig. 3. Shock pattern in the vicinity of the triple point A

Traditional Hugoniot conditions in the perfect gas as well as geometrical (the relation between oblique shock slope and flow deflection angle) were the basic for the further analysis.

$$\tan(\theta) = \frac{2}{\tan(\varphi)} \cdot \frac{M^2 \cdot \sin^2 \varphi - 1}{M^2 \cdot (\gamma + \cos(2\varphi)) + 2}$$

Here φ is the inclination angle (the angle between the shock and the incident flow), θ is the flow deflection angle, M_I is the flow Mach number upstream the given shock. If the quantities of M_I , γ and $\omega_1 = \varphi_1$ are given, then it is possible to determine all flow parameters behind the shock waves and angles of the triple configurations (see, for example, [11] for the full system of equations).

The calculations were performed in a wide range of initial parameters. Flow Mach number varied from 3 to 20, the ratio of the specific heats varied in the range from 1.05 to 1.66, and the incident angle ω_1 – from 10 to 60°.

The dependences between the reflection angle ω_2 and the incidence angle ω_1 for various values of Mach number M_I and the specific heat ratios γ are shown in Fig. 4. The calculations have revealed that the reflection angle becomes negative at large March number and $\gamma < 1.4$. At $\gamma \geq 1.4$ the reflection angle never becomes negative at any Mach numbers (see Fig. 4a).

a)

b)

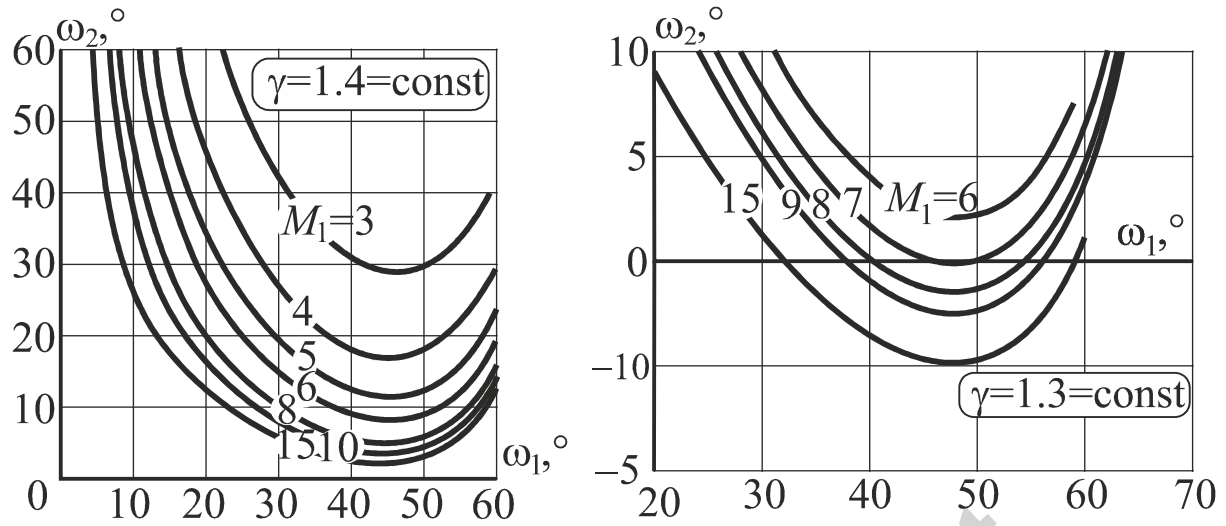


Fig. 4. Relation between the reflected angle ω_2 and incident angle ω_1 at (M_1, ω_1) -plane for various Mach numbers and specific heats ratios $\gamma=1.4$ (a) and $\gamma=1.3$ (b)

Fig. 4 reveals that the curves $\omega_2=f(\omega_1)$ are located only in a positive region of the diagram at constant specific heat ratio $\gamma=1.4$ and any values of M_1 . The similar picture is observed at large values of γ ($\gamma \geq 1.4$). It explains the fact that all triple configurations observed in aerodynamic tunnel experiments form only positive reflection angle because the working gas is air with $\gamma=1.4$.

When the specific heat ratio decreases, for example, to $\gamma=1.3$ (Fig. 4b), the curves shift downward and some curves are located in the negative region of the diagram at large values of Mach numbers. It happens because the gas compressibility increases with γ decreasing and the flow deflection at the reflected wave grows. The flow deflection also grows with the Mach number increasing. Fig. 5 presents the boundaries of the configuration with a negative reflection angle ($\omega_2=0$) in (M_1, ω_1) coordinates for different values of the specific heats ratio. The negative reflection angles are formed inside the regions bordered by those curves.

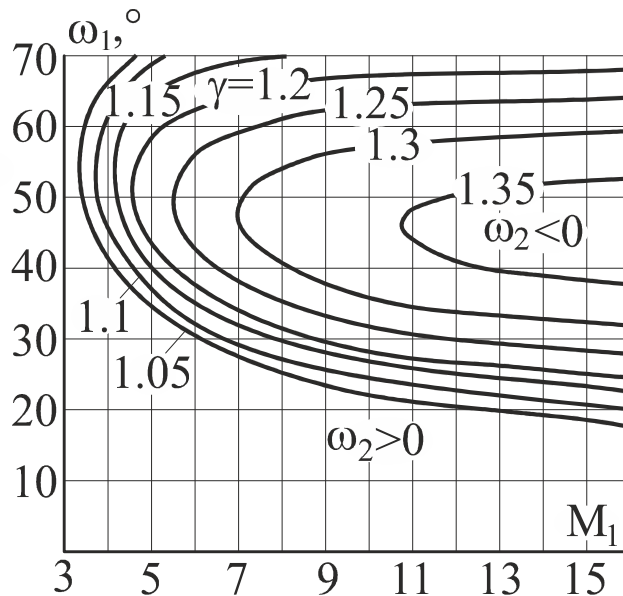


Fig. 5. Boundaries of triple configurations with negative angle ($\omega_2=0$) in (M_1, ω_1) coordinates and various γ 's. Angle ω_2 is negative in the regions inside the corresponding curves

Thus, boundaries and regions of the new configuration with negative reflection angle are found. The configuration with a negative reflection angle is formed when Mach number M_1 of the undisturbed flow exceeds 3 and the specific heats ratio is less than 1.4.

Influence of the Ratio of Gas Specific Heat Ratio on Regular/Mach Transition Conditions

The effect of the value of γ value on the regular/Mach reflection (RR/MR) transition criteria can be also found and regions of double-shock and triple-shock configurations can be constructed as a function of the specific heats ratio and flow Mach numbers.

The arrangement of the shocks in triple configurations depends on three parameters – the free flow Mach number M_1 , the incidence angle ω_1 and the specific heats ratio γ . Fig. 6 illustrates the well-known boundaries between RR and MR in (M_1, ω_1) coordinates for $\gamma=1.4$. It is seen that a dual solution domain exists, where both configurations are possible. This fact explains the so-called hysteresis phenomenon [18, 19]. In case of a very slow transition from the regular reflection to the triple configuration the regular reflection is detained and exists in a region above the angle ω_N (the von Neumann criterion) up (or almost up) to the angle ω_D (the detachment criterion). The opposite slow transition from the Mach reflection to the regular reflection takes place at ω_N boundary line. These conclusions follow from theoretical considerations, but physical experiments reveal that the Mach reflection is more stable in a dual solution domain than the regular one. The hysteresis phenomenon was studied in a lot of works but only for a constant specific heat ratio $\gamma=1.4$. The relation between regular/Mach boundary and the value of γ can give us a complete pattern of the reflection including the new reflection configuration (Fig. 7).

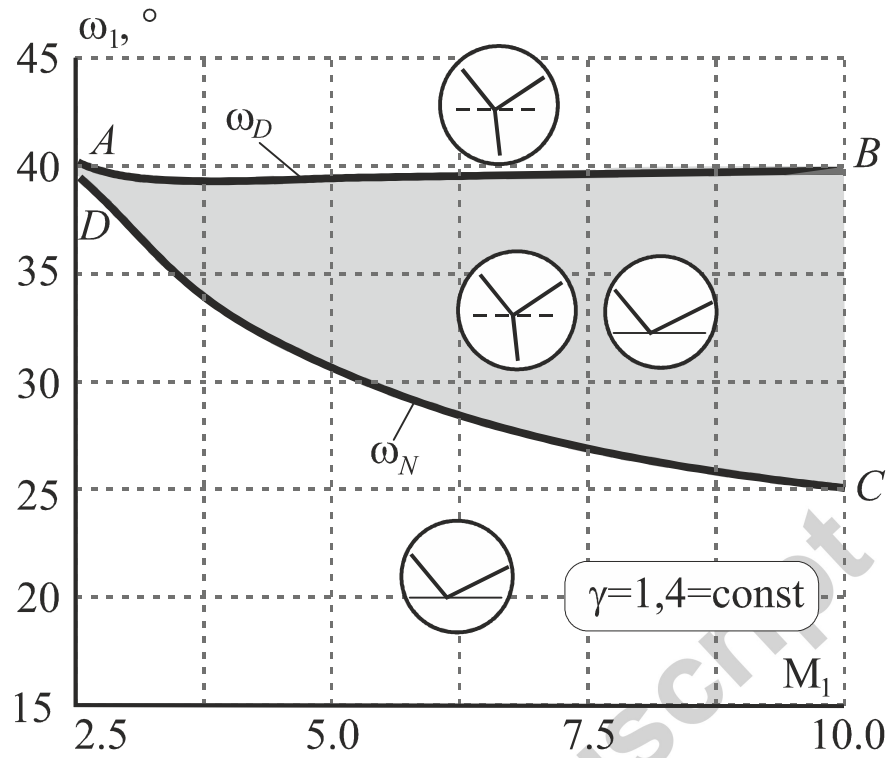


Fig. 6. Transition boundaries according to the two well known criteria (ω_D and ω_N) at the ratio of specific heats $\gamma=1.4$. ABCD is dual solution region, where both reflections are possible

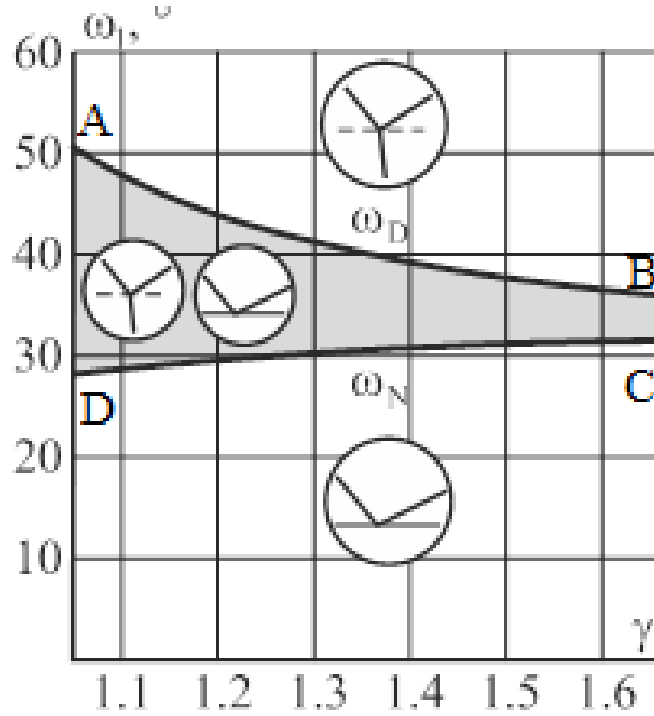


Fig. 7. The transition angles ω_D and ω_N in relation to a specific heats ratio γ for free flow Mach number $M_1=5$. Both regular and irregular reflections are possible in ABCD region

The angle ω_D of RR/MR transition has been found in dependence on the free flow Mach number M_1 and the specific heats ratio γ (RR does not exist at large incidence angles). Critical angle ω_N for the

irregular reflection (MR does not exist below this angle) also has been determined (Fig. 7). The region of double solution is located between these two curves. The effect of the ratio of gas specific heats has been also studied. It is known that the dual solution region for a constant specific heat ratio $\gamma=1.4$ increases rapidly with growth of Mach number [19]. It has been shown that the same relation is correct for the other values of γ . The calculations reveal (Fig.7) that the dual solution region widens with a decrease of the value γ . For example, for a constant $M_1=5$ and the variation in γ from 1.4 to 1.05 leads to the dual solution region increase by 14° .

The Complete Picture of Shock Wave Reflection.

Let us superimpose the RR/MR transition boundaries and the boundary ω_G of the negative reflection angle configuration. Fig. 8 illustrates the complete pattern of all types of reflections for $\gamma=1.2$. The diagram includes all three boundaries of existence: regular reflection (AB), irregular (CD) and the configuration with a negative reflection angle (EGH). The picture reveals that, except for the dual solution region $AGHCD$, where both Mach and regular reflections are possible, there is a new dual solution region BGH . Both RR and MR with a negative reflection angle can exist there. Note, that only one region of double solution exists (see Fig. 6) at $\gamma \geq 1.4$ because the negative reflection angle configuration is not possible then.

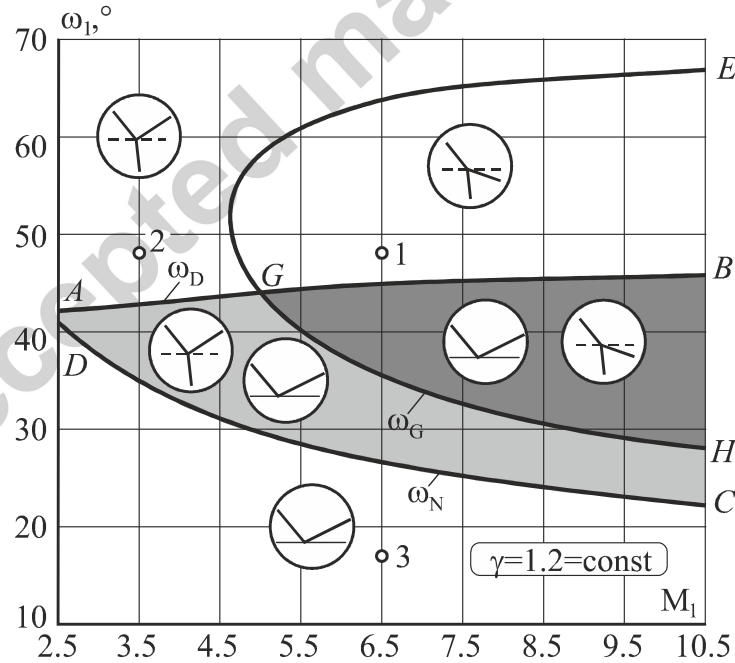


Fig. 8. Transition angles ω_N , ω_D , and ω_G for $\gamma=1.2$. $AGHCD$ is the dual reflection solution domain (regular and irregular reflections are both possible). BGH is the region where both RR and MR with negative reflection angle can exist

Negative Angle Configuration Effect in the Problem of Supersonic Flow in an Air.

What amendments are expected in the flight regimes when a triple configuration with negative reflection angle occurs? The pattern of a flow depends not only on M_1 , ω_1 и γ , but also on some other

conditions. As an example, let us consider a problem of a supersonic inlet problem (Fig. 1b). The triple point theory determines shock disposition in the triple point vicinity. The position of the triple point itself depends on a distance between the wedge and the reflecting surface. The Mach wave cannot be straight along the whole length, it should come to the reflecting surface normally. Since the flow behind that wave is subsonic, downstream conditions, for example some obstacles, affect the triple point position.

The rarefaction wave fan RF emanating from the lower edge of the wedge (Fig. 1b) intersects the reflected wave (the theory of shock / expansion wave interactions is elaborated in [20, 21]). The reflected wave is usually attenuated and becomes convex downwards. Compliance of tests with the triple shock theory was recorded only in vicinity of the triple point at condition that the gas flow behind the reflected shock wave was supersonic, i.e. only for rather strong shock waves. What can be expected in such parameters M_1 , ω_1 and γ that the reflected shock AR is inclined down to the plane or axis of symmetry in the vicinity of the triple point A (Fig. 9a)?

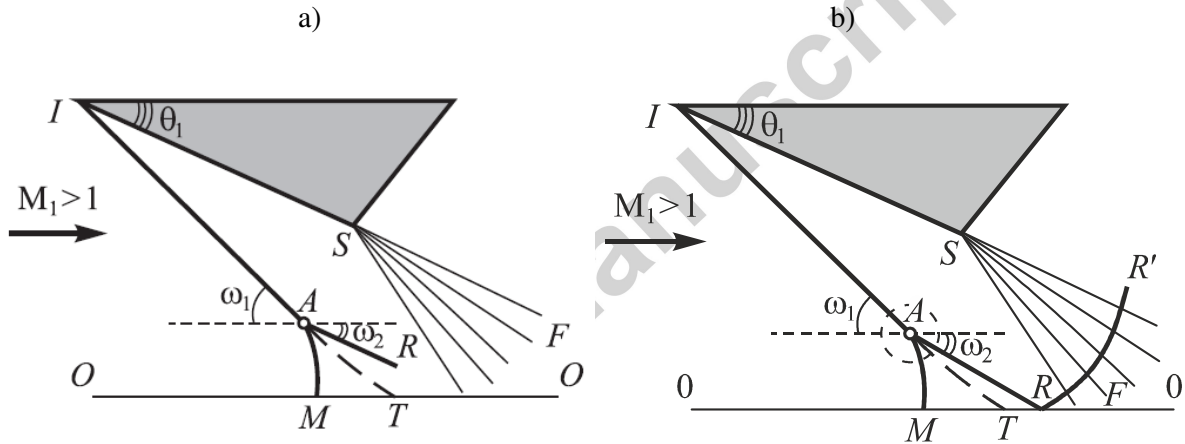


Fig. 9. Supersonic flow in an air inlet: a) configuration of triple shock wave with negative angle of reflection ($\omega_2 < 0$); b) the shock pattern with the enveloped region AMR which can not be realized in a steady flow.

It is obvious that a configuration in which a reflected wave is directed downward intersects the symmetry line, can not be stable. An enveloped region accumulating a gas is formed on the symmetry line then (Fig. 9b). The steadiness of the flow would be then interrupted. Tangential surface should curl into a vortex. New unsteady wave configurations or self-oscillation modes of flowage anticipated. In any event, a configuration with negative angle of reflection leads to the flow mode alteration. This hypothesis was stated in [22].

This statement cannot be verified experimentally using the wind-tunnels that operate with an air at $\gamma=1.4$. Therefore, a complete pattern of the negative angle configuration flow has been studied numerically [14-16]. Computer software Star CCM developed for the mechanics of continuous medium [23] was used. The software uses the Navier-Stokes equations averaging method (RANS - Reynolds-averaged Navier-Stokes). The Spalart-Allmaras model [24] was used for description of turbulent stresses. A problem of shock wave reflection from a symmetry plane at the interaction of oblique shocks generated

by two symmetrical wedges in a supersonic flow (Fig. 1b) was assumed as a model problem. Calculations were performed at a different values of the wedges angles ($\omega_1=10-50^\circ$) and for a wide range of flow parameters ($M_1=3-10$, $\gamma=1.1-1.4$).

The main attention paid to the calculation in the region of a negative reflection angle (*EGB* region in Fig. 8). The study of this region is complicated because of the influence of the geometrical characteristics of the downstream flow. First of all, the rarefaction fan formed at the lower edge of the wedge affects the shock wave pattern (Fig. 1b). To obtain a triple shock configuration with a negative angle of reflection it is necessary to find such a geometry at which the rarefaction wave intersects the reflected shock only; that was done by the variation of the distance between the bottom edge of the wedge and the symmetry line. As a result, the solution was obtained in the region of the negative angle of reflection for the stationary supersonic flow [15].

The numerical study has revealed that there is no steady solution in this region. An unstable configuration climbing upstream the flow is formed. Its form depends on the previous type of reflection. It was studied, for example, what configuration is formed at flow parameters corresponding to the point 1 (Fig. 8). If the transition to the point 1 took place from the steady Mach reflection at the parameters corresponding to point 2, Mach reflection was preserved, and the configuration shown in Fig. 10 was formed.

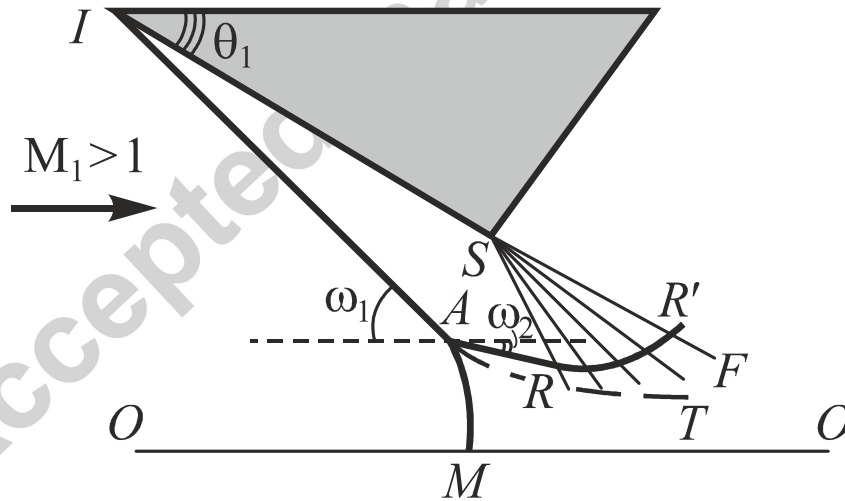


Fig. 10. The scheme of new configuration with a negative reflection angle and a kink in the reflected shock.

Interaction between reflected shock and rarefaction waves of the opposite direction [20] causes deviation of the reflected shock towards the positive slope angle. However, in spite of the fact that in this case the reflected shock does not intersect the symmetry line, the resulted configuration is unstable. The solution did not converge and the triple point climbed upstream the flow. This fact needs further studies.

At the continuous variation of flow parameters to the point 1 (Fig. 8) from the regular reflection which occurs at the parameters corresponding to the point 3, a configuration with the second triple point on the reflected wave appears (so-called double Mach reflection, DMR, Fig. 11). The reflected wave

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1} (M_1^2 \sin^2 \omega_1 - 1)$$

Here p_1 is the pressure at nozzle lips, p_2 is external pressure.

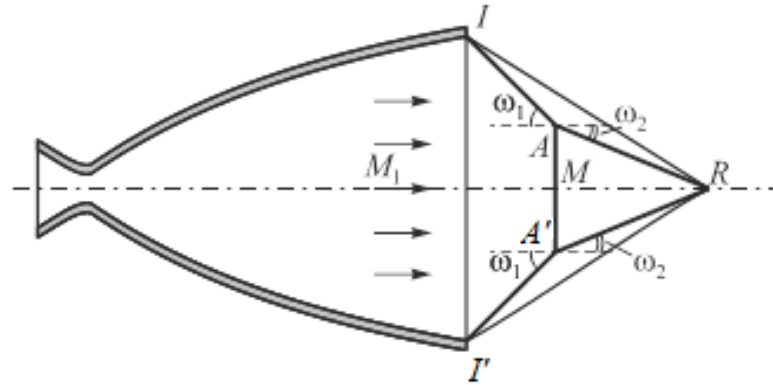


Fig. 12. Triple configuration with negative reflection angle ($\omega_2 < 0$) in an overexpanded planejet flow. IA, IA' – incident shock waves, AA' – Mach bridge, AR, A'R – reflected wave. AT, A'T – slipstreams.

It was shown that the transition to flow parameters corresponding to triple configurations with negative reflection angles at a flow around two wedges leads to disappearance of the steady numerical solution and to the oscillations of the shock pattern of the flow. This effect of the inlet flow pattern alteration should be anticipated at high velocities of aircraft or spacecraft in the air, when non-equilibrium processes occur and therefore the specific heats ratio decreases (becomes less than 1.4). The same effect is possible at spacecraft flight in Mars atmosphere, which consists of carbon dioxide (γ is much less than 1.4).

This effect also can be realized in a supersonic jet out a nozzle (Fig. 1a) due to a low value of the specific heats ratio of conventional fuels. Fig. 1a and Fig. 1b show that there is a similarity of shock patterns between a flow in an air inlet and a flow in an overexpanded supersonic jet. It has been shown that if a reflected shock in an air inlet directed at negative angle ω_2 intersects symmetry line OO_1 a gas is accumulated in an enveloped region ARM . The flow is closed in this region and the Mach wave begins to climb upstream the flow, an unsteady double Mach reflection would be formed. The similar process is anticipated in an overexpanded jet with a negative angle of reflected shock (Fig. 12). Supersonic gas flows should collide in the closed region behind the reflected shocks AR and $A'R$. Vortices would form in ARA' region, and the Mach stem would climb upstream the flow up to the nozzle entrance. This can might cause the flow separation in the nozzle and start a self-oscillation process. Thus, the results obtained can be useful for the prediction of in-flight accidents in supersonic jet propulsion engines as well as in the atmospheres of other planets.

Acknowledgment

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Highlights

- New pattern of the triple-shock configuration is studied
- Reflected shock is inclined downwards the initial flow direction
- Instability of this configuration leads to a radical change of flow pattern
- This instability influences the Mach reflection at small specific heats ratios